



CIVIC Pulse

Smarter Reporting and faster resolution.

CAPSTONE PROJECT

IIT PATNA

BATCH 4 • 2026

GROUP 11



IIT PATNA

An AI-powered platform that lets residents report civic issues with a **photo**, a **comment**, and a **location** — and lets municipal teams triage, route, and resolve them without the guesswork.



Report



AI Triage



Route



Track



Resolve & Rate

THE PROBLEM



Fragmented channels — issues reported by phone, in person, or on social media, none tracked systematically.



No visibility — residents can't tell if a report has been seen, assigned, or resolved.



Duplicate effort — the same pothole gets reported many times, creating redundant tickets.



Slow manual triage — categorizing, prioritizing, and routing issues by hand delays response.



No data-driven oversight — no aggregated view of volume, resolution time, or hotspots.

OUR SOLUTION

CivicPulse turns a photo and a comment into a fully triaged, trackable service ticket. AI reads the image and text together, classifies the issue, checks for duplicates, and routes it to the right department — with residents and staff updated at every stage.



Report Issue



AI Classifies



Routed / Kanban



Resolved & Rated



RESIDENT APP

- Report via phone, email, or anonymously
- Photo + free-text comment per issue
- Auto GPS capture with PIN code, area, city, state
- Optional nearby-landmark field
- Voice-note input for low-literacy / first-time users
- Closure confirmation with resolution photo & staff notes

Multilingual UI (English, Hindi + regional) — phased rollout



AI & DEDUPLICATION

- Joint image + text analysis, not either alone
- Automatic issue-type categorization
- Automatic responsible-department identification
- Priority scoring against defined SLA rules
- Matches new reports to open issues by area + type
- Merges duplicates into one ticket, not many

Powered by Claude API — image + text reasoning



OPS & KANBAN BOARD

- Custom Kanban board across lifecycle stages
- Drag-and-drop / action-based status transitions
- Filter by area, department, priority, status
- 3-tier SLA auto-escalation with audit trail
- Email, SMS & Telegram notifications
- Analytics on volume, resolution rate & hotspots

Realtime board updates via Supabase

01 TICKET LIFECYCLE

1

CREATION
Validated, non-duplicate report

2

ASSIGNMENT
Routed by AI categorization

3

IN PROGRESS
Status updates as work proceeds

4

RESOLVED
Proof image + comment attached

5

CLOSED
Verified, resident notified

6

REOPENED
If resident rates it unsatisfactory

AUTO-ESCALATION



SLA TIERS

TIER 1

Department user has until the SLA deadline to move the ticket to In Progress or Resolved.

TIER 2

On breach, supervisor is notified and the ticket is flagged "Overdue" on the board.

TIER 3

Still unresolved after a second window → escalates to department head.

KANBAN BOARD — Illustrative layout

CREATED	ASSIGNED	IN PROGRESS	RESOLVED	OVERDUE
Streetlight out Sector 12	Water leakage PWD Dept	Pothole — Main Rd Roads Dept	Broken drain cover Closed pending rating	Streetlight, Ward 7 Tier 2 — Supervisor notified
Garbage pile-up Ward 4		Damaged railing Sector 9		

Filterable by area • department • priority • status

AI CAPABILITIES

- Image + text understanding, jointly
- Issue-type categorization
- Department auto-routing
- SLA-based priority scoring
- Duplicate / image-signature matching
- RAG chatbot on historical issue data
- Speech-to-text on voice notes
- Auto-translation for UI & notifications

RAG chatbot answers resident & staff questions, grounded in past reports and resolutions.



TECHNOLOGY STACK

LAYER	TECHNOLOGY	PURPOSE
LLM / AI	Claude Code, Claude API	Image + text analysis, categorization, routing, SLA priority, RAG chat
Backend & DB	Supabase	Postgres, auth, file/image storage, realtime board, auditing
Frontend	Claude Code, etc.	dashboards, analytics
Notifications	Email, SMS, Telegram	Status updates & SLA escalation alerts
Voice & Language	Speech-to-text + translation	Voice-note transcription, multilingual UI (optional)

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OUR TEAM



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Yogi Vasoya

See it. Report it. Track it resolved.